

CARROLL COUNTY REGIONAL, MD, USA

WMO: 720699

Lat: 39.608N

Lon: 77.008W

Elev: 789

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 10-19

WBAN: 03724

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	13.5	-7.7	3.8	16.1	-1.8	5.1	19.7	23.9	32.9	20.8	28.4	7.5	290	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	93.2	75.3	90.9	74.5	89.6	74.2	80.2	88.0	77.8	87.1	76.3	85.2	6.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.7	153.3	84.9	75.0	135.2	82.9	73.1	126.6	80.8	44.5	88.7	41.7	88.5	40.2	84.0	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.7	17.2	15.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			3.8	83.5	6.0	3.4	-0.5	85.9	-4.0	87.9	-7.4	89.7	-11.7	92.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	31.2	35.9	43.1	55.6	66.4	73.8	78.9	75.8	69.4	57.9	45.5	38.0
	DBStd	18.28	10.20	11.65	10.32	9.50	8.56	5.74	5.12	4.99	7.01	8.40	8.30	9.19
	HDD50	1906	585	415	269	46	1	0	0	0	21	180	389	
	HDD65	4667	1046	817	682	312	92	6	0	37	252	585	838	
	CDD50	4122	3	19	54	214	510	715	895	800	583	267	46	16
	CDD65	1410	0	1	1	29	137	272	430	336	170	33	1	0
	CDH74	11981	0	7	14	256	1128	2317	4159	2722	1184	193	1	0
	CDH80	4584	0	0	0	69	390	840	1836	1012	393	44	0	0
Wind	WSAvg	5.7	7.1	6.9	7.2	7.1	5.3	5.0	4.1	3.6	4.0	5.7	6.4	6.3
Precipitation	PrecAvg	41.5	3.0	2.5	3.7	3.5	4.2	3.8	3.9	3.5	3.7	3.1	3.3	3.3
	PrecMax	62.3	8.0	5.4	7.6	7.7	9.0	18.2	9.4	8.8	17.6	8.3	7.3	7.7
	PrecMin	29.7	0.6	0.4	1.3	0.9	0.5	0.4	0.4	1.0	0.8	0.1	0.4	0.3
	PrecStd	8.1	1.5	1.3	1.6	1.7	1.9	3.0	2.1	2.0	3.3	1.8	1.7	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	72.3	75.3	87.7	91.3	93.5	99.0	94.9	91.3	84.3	72.2	66.0
		MCWB	56.2	60.9	58.3	63.6	74.2	74.3	77.8	75.4	72.4	71.2	62.1	63.2
	2%	DB	55.0	63.5	70.2	81.3	88.3	90.8	94.9	91.1	88.5	79.4	66.1	62.6
		MCWB	50.0	54.3	57.9	63.1	73.3	74.6	77.9	73.6	72.4	67.1	58.4	59.8
	5%	DB	50.4	57.2	64.1	75.3	84.1	88.2	91.3	89.6	84.3	74.7	62.8	55.4
		MCWB	46.1	50.1	54.7	60.3	70.5	73.5	75.7	72.8	71.2	65.9	55.0	51.9
	10%	DB	45.4	52.1	60.5	71.7	81.1	84.3	89.8	85.9	81.2	71.8	57.5	51.7
		MCWB	41.2	45.8	51.5	59.3	68.2	71.6	75.7	71.4	68.7	63.7	50.3	48.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.9	62.9	64.1	70.3	81.2	81.1	85.3	81.9	77.4	73.7	64.2	63.7
		MCDB	59.6	71.4	70.1	79.7	85.8	86.9	91.1	88.3	87.3	79.4	67.3	65.7
	2%	WB	51.1	57.3	60.3	66.6	74.8	77.7	81.7	77.4	75.3	71.1	60.4	59.5
		MCDB	54.0	60.6	67.2	74.8	84.3	85.7	90.1	87.0	83.9	77.1	64.4	62.2
	5%	WB	46.0	50.8	56.8	63.9	72.7	75.5	78.7	75.5	73.4	68.1	57.1	53.8
		MCDB	48.9	56.8	62.8	72.5	81.8	84.3	87.9	84.5	81.6	73.6	62.0	55.5
	10%	WB	41.7	45.9	52.2	60.9	70.5	74.0	77.1	74.0	71.5	64.5	53.3	47.4
		MCDB	45.4	49.3	58.9	69.3	78.9	82.4	86.8	81.4	77.8	69.0	56.5	49.2
Mean Daily Temperature Range	5% DB	MDBR	13.8	15.6	17.1	20.5	19.0	19.4	19.1	18.7	18.4	17.0	15.5	12.5
		MCDBR	20.0	23.2	23.5	25.4	23.3	22.2	21.5	21.5	22.3	20.9	20.0	17.0
		MCWBR	17.9	16.5	14.9	12.9	11.6	10.2	8.8	9.5	10.0	11.1	13.1	15.3
	5% WB	MCDBR	18.4	21.1	22.3	23.6	22.0	19.6	19.3	18.8	18.1	16.3	16.5	16.3
		MCWBR	17.0	17.1	15.7	13.8	13.0	12.3	11.0	10.1	9.3	10.1	13.4	16.4
Clear-Sky Solar Irradiance	taub	0.304	0.315	0.353	0.394	0.430	0.492	0.507	0.501	0.420	0.357	0.318	0.302	
	taud	2.447	2.395	2.369	2.276	2.206	2.070	2.044	2.045	2.245	2.427	2.498	2.488	
	Ebn at Noon	275	287	286	279	270	253	247	245	260	268	268	267	
	Edh at Noon	25	31	35	41	45	52	53	51	40	30	24	23	
All-Sky Solar Radiation	RadAvg	612	891	1182	1524	1701	1888	1877	1660	1355	990	707	534	
	RadStd	52	106	80	128	195	147	101	90	171	110	61	56	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (73 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+121	

Nomenclature: See separate page